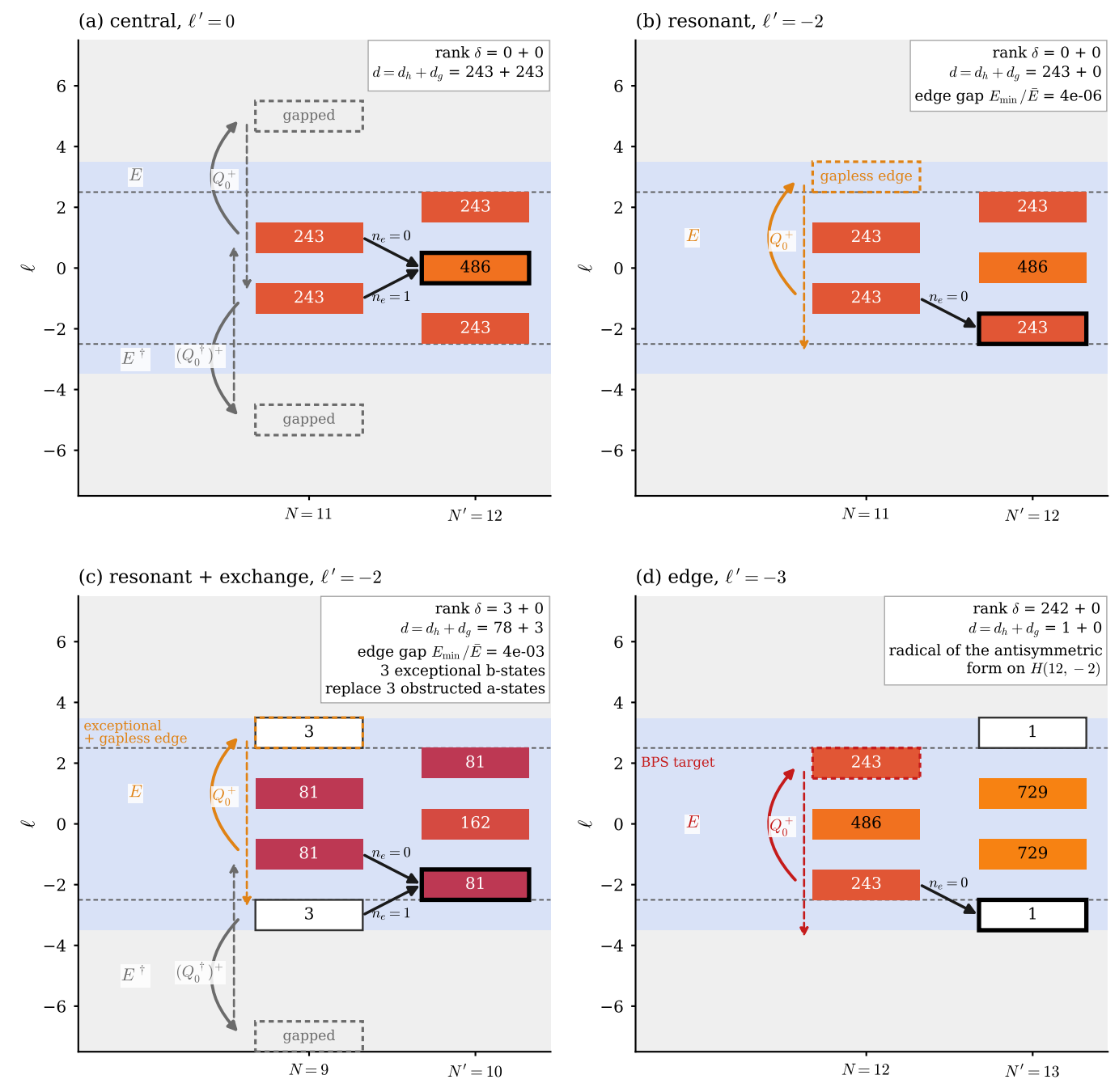
(a) central, $\ell' = 0$
rank δ = 0 + 0
$d = d_h + d_g$ = 243 + 243
gapped
E
Q_0^+
243
243
$n_e = 0$
486
$n_e = 1$
243
243
$E^\dagger$
$(Q_0^\dagger)^+$
gapped
ℓ
$N = 11$
$N' = 12$
(b) resonant, $\ell' = -2$
rank δ = 0 + 0
$d = d_h + d_g$ = 243 + 0
edge gap $E_{\min}/\bar{E}$ = 4e-06
gapless edge
E
Q_0^+
243
243
$n_e = 0$
243
486
243
ℓ
$N = 11$
$N' = 12$
(c) resonant + exchange, $\ell' = -2$
rank δ = 3 + 0
$d = d_h + d_g$ = 78 + 3
edge gap $E_{\min}/\bar{E}$ = 4e-03
3 exceptional b-states
replace 3 obstructed a-states
exceptional
+ gapless edge
3
E
Q_0^+
81
81
$n_e = 0$
162
81
81
$n_e = 1$
3
$E^\dagger$
$(Q_0^\dagger)^+$
gapped
ℓ
$N = 9$
$N' = 10$
(d) edge, $\ell' = -3$
rank δ = 242 + 0
$d = d_h + d_g$ = 1 + 0
radical of the antisymmetric
form on $H(12, -2)$
BPS target
243
E
Q_0^+
729
486
$n_e = 0$
729
243
1
1
ℓ
$N = 12$
$N' = 13$
jump (walk step)
dressing, target outside window
dressing, target = window edge
dressing, target inside window
return Q_0^+ / $(Q_0^\dagger)^+$